

Guideline Title: Ventricular Fibrillation and Pulseless Ventricular Tachycardia

Guideline Number: GUID-3203-C007

Issue date: 01/01/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** Successful treatment of ventricular fibrillation and pulseless ventricular tachycardia with a return of spontaneous circulation.
- II. **DEFINITIONS:**
 - a. VF – ventricular fibrillation
 - b. VT – ventricular tachycardia
 - c. ROSC – return of spontaneous circulation
 - d. CPR – Cardiopulmonary resuscitation
- III. **DEPARTMENT GUIDELINE:**
 - a. Initiate CPR, administer oxygen, apply ETCO₂ monitor if advanced airway present, and maintain ETCO₂ > 10 mmHg.
 - i. If available, an FDA approved cardiac compression/mechanical CPR device may be applied and utilized per manufacturer's instructions
 - b. Apply cardiac monitor, confirm VF or VT, and prepare for defibrillation.
 - c. *Defibrillate per GUID3203-PR008* for persistent VF/VT 120-200 J for biphasic. Resume CPR immediately after defibrillation.
 - i. For refractory VF/VT after >3 standard defibrillation attempts, consider Dual Sequential Defibrillation (DSED) if a second monitor is available (*see GUID3203-PR008*)
 - d. Establish IV/IO access (consider 2 sites).
 - e. Approximately every 2 minutes, reassess cardiac rhythm and assess ROSC. If no ROSC and a shockable rhythm, repeat defibrillation at 360 J. Continue compressions.
 - f. Administer **Epinephrine (1:10,000) 1 mg IV/IO** push, repeat every 3-5 minutes.
 - g. Secure advanced airway and ventilate per *GUID3203-PR020 Endotracheal Intubation* with 100% oxygen
 - h. Administer one of the following antidysrhythmic:
 - i. **Amiodarone 300 mg IV/IO**, may repeat 150 mg in 3-5 minutes **-OR-**
 - ii. **Lidocaine 1-1.5 mg/kg IV/IO** may repeat every 5-10 minutes at 0.5-0.75 mg/kg (max. dose 3 mg/kg).

Guideline Title: Ventricular Fibrillation and Pulseless Ventricular Tachycardia

Guideline Number: GUID-3203-C007

- i. Consider **Magnesium sulfate 1-2 grams IV/IO** over 2 minutes for Torsades de Pointes or known hypomagnesemia (chronic alcohol use or diuretic use).
- j. Consider continuous IV/IO infusion of the antidysrhythmic that proved to be effective in converting rhythm.
 - i. **Amiodarone infusion at 1 mg/min x 6 hours**
 - ii. **Lidocaine infusion at 1-2 mg/min** titrate up to **4mg/min** for effect.
- k. Consider **Sodium Bicarbonate 8.4% 50 mEq IV/IO** for suspected hyperkalemia in dialysis patients or tricyclic antidepressant overdoses.
- l. If ROSC, refer to *GUID3203-C009 - Targeted Temperature Management*.
- m. Always consider the latest recommendations from the American Heart Association.

IV. DOCUMENTATION:

- a. EMS Charts

V. REFERENCES:

- a. American Heart Association, 2015.